

BREEZE: Physics-Verified LLM Recipe Generation for Few-Shot Condition Monitoring

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ABSTRACT

Scarce labelled faults make it costly to train and select a signal generator for every machine and operating regime. We introduce BREEZE, a training-free workflow that separates LLM recipe proposal, seeded deterministic rendering, and physical admission calibrated exclusively on the real training split. The verifier records each gate decision, rejects failed candidates unchanged, and uses structured failure reports for bounded resampling. On file-disjoint PU, LLM recipes outperform matched rule and random open-loop recipes for Accuracy and Macro-F1 at 5, 10, and 25 real windows per class; all 12 registered comparisons pass Holm correction ($q < 0.001$, 20 seeds), as do all 72 provenance-valid CWRU comparisons across within-load0 and held-out loads 1–3 (40 seeds). On case/run-disjoint Berkeley milling, 15/18 comparisons pass, including all 12 against non-structured baselines; its lower-shot LLM–rule advantage, class-conditional diagnostics, and six failed PU cross-condition attempts delimit the evidence. A pass states train-calibrated admissibility; these results establish auditable, physics-verified LLM-guided admission for the stated few-shot protocols, outside claims of formal physical validity, zero-data diagnosis, or universal generator superiority.

1. Introduction

Industrial fault diagnosis faces a structural data imbalance. Healthy operation is abundant, whereas labelled faults are scarce, expensive to seed, and rarely observed under every load and speed. New machines, sensor layouts, and operating regimes therefore offer only a few examples of the events that a diagnostic model must recognize.

Existing augmentation routes trade modelling capacity for target-data cost. Conventional transformations are inexpensive, while VAEs, GANs, and diffusion models can learn richer signal distributions through target-data training and model selection (Li, Zhang, Ding and Sun, 2020; Zhao, Liu, Gu, Sun, Wang, Wei and Zhang, 2020; Zhang, Ye, Wang and Habetler, 2021; Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville and Bengio, 2014; Kingma and Welling, 2014; Ho, Jain and Abbeel, 2020). This investment is practical for a large stable corpus, but becomes costly when each new deployment supplies only a few labelled faults.

LLM-authored recipes create an inference-time alternative to target-specific generator training. An LLM can express machine metadata, fault class, and signal structure as a compact recipe, which a numerical renderer executes as a waveform. This separation turns the language model into a proposal source and keeps signal construction explicit.

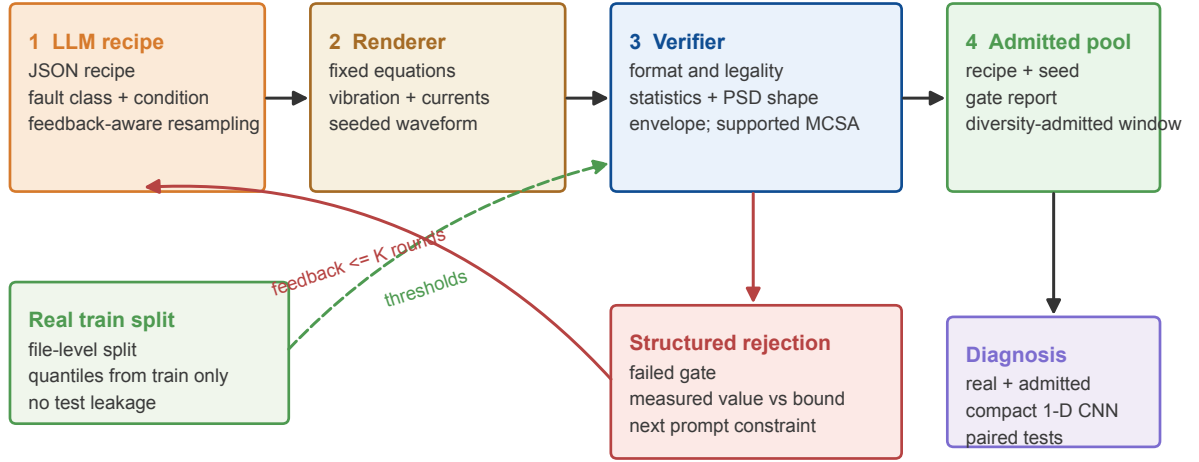
The recipe interface also creates a reliability gap between valid syntax and supported signal evidence. A well-formed recipe can still yield unsupported amplitudes, implausible spectral allocation, missing fault-frequency evidence, or repeated samples. Prompt instructions provide no deterministic basis for deciding which candidates should enter a diagnostic training pool.

BREEZE closes this gap through train-calibrated admission. The LLM proposes a JSON recipe, a fixed renderer maps the recipe and seed to a waveform, and a deterministic verifier evaluates gates calibrated from the corresponding outer training split. Each rejection report identifies measured violations and conditions a bounded resampling round, while the held-out test split remains isolated from recipe construction and admission. Figure 1 summarizes this closed loop.

Three evaluation questions organize the evidence. First, does the LLM contribute beyond random or engineer-written recipes under a matched renderer? Second, do admitted signals exhibit train-supported time-domain, spectral, envelope, and diversity evidence? Third, does the resulting pool provide stable diagnostic utility under file-, load-, or experiment-disjoint few-shot protocols? Paired source ablations, class-conditional diagnostics, leakage-resistant splits, and corrected statistical tests answer these questions together.

The paper makes four contributions:

a

Closed-loop physics-verified admission

No target-data generator optimization and no waveform repair: rejected candidates remain rejected.

Figure 1: The BREEZE workflow. The LLM proposes a structured recipe, a fixed seeded renderer constructs vibration and, when available, current channels, and a verifier calibrated exclusively on the real training split admits or rejects the candidate. A rejection becomes structured feedback for at most K further rounds. Rejected signals are not repaired.

1. **Separate** LLM time-series generation into a recipe–renderer–verifier system, enabling independent auditing of language proposals, numerical signal construction, and train-calibrated admission.
2. **Integrate** mixed physical gates—format legality, robust signal statistics, Welch-PSD shape, envelope-spectrum evidence, reliability-gated current evidence, and diversity—into a bounded rejection/resampling loop without training a target-data generator.
3. **Establish** the contribution of LLM recipes under matched few-shot protocols on three public datasets: PU, CWRU, and Berkeley milling. The registered evidence comprises 20 PU seeds and 40 seeds for CWRU and Berkeley, with family-wise Holm correction.
4. **Report** negative evidence as part of the method boundary: the full six-stage PU leave-one-condition-out (LOCO) failure chain, process-metadata confounding in UMich milling, and a stopped MU-TCM inner-validation line.

The evidence supports a deliberately bounded admission claim. BREEZE shows that accepted candidates satisfy declared train-supported gates and improve the stated few-shot tasks under their frozen protocols. It does not establish zero-real-data operation, universal cross-condition morphology transfer, formal physical correctness, or superiority to every learned generator.

2. Related work and positioning

2.1. Time-series generation for diagnosis

Trainable models dominate general time-series generation. Their architectures include recurrent conditional GANs, TimeGAN, transformer GANs, variational models, and diffusion models (Esteban, Hyland and Rätsch, 2017; Yoon, Jarrett and van der Schaar, 2019; Li, Metsis, Wang and Ngu, 2022a; Li, Ngu and Metsis, 2022b; Liao, Ni, Szpruch, Wiese, Sabate-Vidales and Xiao, 2020; Desai, Freeman, Wang and Beaver, 2021; Nichol and Dhariwal, 2021; Kong, Ping, Huang, Zhao and Catanzaro, 2021; Tashiro, Song, Song and Ermon, 2021; Alcaraz and Strodthoff, 2022; Rasul,

Table 1

Positioning against representative synthetic-signal systems. “Target training” denotes optimization of a generator on the target signal corpus.

Method family	Generative mechanism	Target training	Physical control	Primary evidence pattern
DFE-GAN / SGAN-DDS	Learned adversarial generator plus filtering or sampling	Yes	Learned features and spectrum guidance	Fidelity, diversity, and diagnosis
Physics-informed GAN / diffusion	Learned generator with physical priors or condition embedding	Yes	Training loss or architecture	Mechanism, multidomain fidelity, diagnosis
BearGen	LLM descriptions condition a learned diffusion model	Yes	Description guidance and learned denoising	Eight bearing datasets, quality, few-shot, cost
BREEZE	LLM JSON recipe plus fixed seeded renderer	No	Train-calibrated admission and rejection	Source ablation, gate reports, physical distances, paired diagnosis

Seward, Schuster and Vollgraf, 2021). Machinery-specific variants address scarce or imbalanced signals through sparsity constraints, deep feature-enhanced filtering, frequency learning, spectrum-guided sampling, and FillingGAN (Ma, Ding, Wang, Wang, Ma and Lu, 2021; Liu, Jiang, Wu and Li, 2022; Ko, Lee, Kim, Kim and Youn, 2023; Kim, Ko, Lee, Kim, Jung and Youn, 2024; Lin, Kong, Zhao, Han, Dong, Liu and Chu, 2025). BREEZE differs by evaluating structured recipe generation at an operating point without target-generator optimization.

2.2. Physics-informed generation

Physics-informed generation embeds equations, mechanisms, or physical penalties in a trainable model (Raissi, Perdikaris and Karniadakis, 2019). Recent rotating-machinery GAN and diffusion systems encode impact response, speed modulation, multimodal features, or condition embeddings (Gao, Xu, Zhang, Fang and Tan, 2025; Liu, Han, Chen, Ding, Zhang, Shen, Wu, Zhang and Li, 2026; Guo, Xu, Zheng, Xie and Wang, 2025). These models can learn their physical objectives through target-data optimization, followed by generator selection.

Generated-data filtering provides a complementary control mechanism. DFE-GAN and SGAN-DDS show that selection and diversity control affect the final pool (Liu et al., 2022; Kim et al., 2024). BREEZE differs by placing mixed bearing-diagnostic gates, train-only calibration, structured feedback, and per-sample audit records inside deterministic inference-time admission.

2.3. LLMs and machinery signals

LLMs provide emerging interfaces to time-series and machinery knowledge. They have been reprogrammed for forecasting without task-specific training, used as numerical forecasters, and supplied with machinery priors for bearing health management (Jin, Wang, Ma, Chu, Zhang, Shi, Chen, Liang, Li, Pan and Wen, 2024; Gruver, Finzi, Qiu and Wilson, 2023; Peng, Liu, Du, Gao and Wang, 2025). BearGen is the closest signal-generation study: an LLM produces signal descriptions that condition a diffusion generator trained across multiple bearing datasets (Lee, Jeon, Kim and Kim, 2026). BREEZE differs by asking the LLM for a structured numerical recipe and executing it with a fixed renderer whose target-data path has no learned weights.

2.4. Diagnostic observables used as gates

Established diagnostic observables expose complementary aspects of rolling-element faults. Envelope analysis, spectral kurtosis, and the fast kurtogram reveal stochastic or pseudo-cyclostationary impacts (Randall and Antoni, 2011; Antoni, 2006, 2007), while Welch PSD estimation represents spectral shape (Welch, 1967). Motor-current sidebands add bearing evidence when the real training data support class separability (Schoen, Habetler, Kamran and Bartheld, 1995). BREEZE differs by converting these observables into explicit, train-calibrated admission gates.

3. Problem formulation and framework

3.1. Train-calibrated admission

Reliable admission requires a decision rule calibrated entirely within the outer training split. Let $\mathcal{D}_{\text{tr}} = \{(x_i, y_i, m_i)\}_{i=1}^n$ denote its real windows, class labels, and available machine metadata. For requested class y and condition m , the LLM proposes recipe r_t at feedback round t . The following mapping makes recipe execution and local randomness explicit:

$$\tilde{x}_{t,s} = R(r_t, m; s), \quad (1)$$

where the seeded renderer R archives s with the recipe. The verifier then maps the rendered waveform to both an admission decision and its evidence report:

$$(a_{t,s}, q_{t,s}) = V(\tilde{x}_{t,s}, y, m; \mathcal{D}_{\text{tr}}), \quad a_{t,s} \in \{0, 1\}. \quad (2)$$

Equation (2) exposes the decision $a_{t,s}$ and the diagnostic report $q_{t,s}$ as separate outputs. Renderer inputs, gate calibration, feedback, and pool selection all remain confined to the outer training split.

Admission must represent interval support, one-sided evidence, healthy suppression, and diversity in one decision. For deterministic feature f_j , class-supported bounds $[l_{j,y,m}, u_{j,y,m}]$, healthy fault-prominence score p_f , physical embedding $e(x)$, and current admitted set S_y , the following system shows the four predicate types:

$$\begin{aligned} l_{j,y,m} &\leq f_j(x) \leq u_{j,y,m} && \text{(interval),} \\ f_j(x) &\leq u_{j,y,m} \text{ or } f_j(x) \geq l_{j,y,m} && \text{(one-sided evidence),} \\ p_f(x) &\leq u_{f,\text{healthy}} && \text{(healthy suppression),} \\ \min_{z \in S_y} \|e(x) - e(z)\|_2 &\geq \delta_y && \text{(diversity).} \end{aligned} \quad (3)$$

The intersection of the active predicates defines the admissible set. Membership therefore records compliance with the configured gates under their training calibration.

3.2. Responsibility boundaries

Attributing an observed gain requires explicit boundaries between proposal, waveform construction, admission, and diagnosis. Figure 2 assigns one component to each responsibility. The LLM selects recipe fields, the renderer enforces the signal equations, the verifier measures compatibility with train-supported evidence, and the downstream classifier measures utility. Matched source comparisons can then isolate the recipe source while holding the other components fixed.

The structured recipe exposes every LLM-controlled degree of freedom. It records shaft and fault frequencies, target RMS, periodic-impact rate, amplitude, decay, resonance, slip jitter, amplitude variation, load-zone modulation, background components, random machine transients, eight vibration-band weights, and optional two-phase current parameters. The initial prompt supplies class, condition metadata, bearing kinematics, a training-derived exemplar description, and the JSON schema. The feedback prompt adds the previous recipe, each measured gate violation, its supported bound, and a request for proportional parameter change. Complete prompt builders and schemas are preserved in `src/llm.py`.

3.3. Seeded renderer

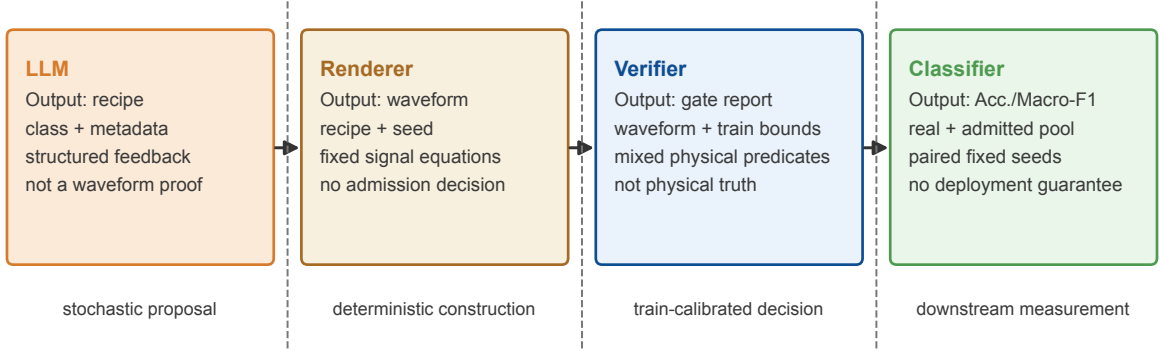
A fixed renderer is needed to compare recipe sources under identical waveform construction. For PU vibration, the renderer combines deterministic background harmonics, Poisson machine transients, a jittered train of resonant impacts, and band-shaped noise. The equations expose the source terms and the link between impact interval and requested fault frequency:

$$x_v(t) = \sum_q A_q \sin(2\pi f_q t + \phi_q) + \sum_{\ell} b_{\ell} h(t - u_{\ell}) + \sum_k a_k h(t - t_k) + \epsilon_b(t), \quad (4)$$

$$h(\tau) = \mathbf{1}_{\tau \geq 0} \exp(-\tau/\tau_d) \sin(2\pi f_r \tau + \phi), \quad (5)$$

$$t_{k+1} - t_k = f_f^{-1} (1 + \xi_k). \quad (6)$$

a Recipe-renderer-verifier responsibility boundary



Only the highlighted component owns each output; evidence is not transferred across responsibility boundaries.

Figure 2: Component responsibility boundary. The LLM selects recipe parameters; the renderer alone constructs waveforms; the verifier admits or rejects without repair; and the classifier is used only for downstream evaluation. Every formal source comparison holds renderer and classifier fixed.

Here u_ℓ follows a Poisson background process, f_f is the requested fault frequency, and ξ_k is bounded recipe jitter. Inner-race impact amplitudes may be modulated at shaft frequency. The renderer shapes additive noise over the fixed eight-band grid and applies the same deterministic target- RMS normalization to every recipe source before admission. Both operations belong to R and precede the gate decision.

When two phase currents are available, the following equation makes carrier, fault modulation, harmonics, and channel noise explicit for channel p :

$$i_p(t) = A_p \left[1 + \beta \cos(2\pi f_f t + \psi) \right] \left[\sin(2\pi f_e t + \phi_p) + \sum_{h \in \{3,5,7\}} \rho_h \sin(2\pi h f_e t + \phi_{p,h}) \right] + \eta_p(t), \quad (7)$$

where f_e is supply frequency. The code and archived seed make $R(r, m; s)$ deterministic.

Berkeley requires a dataset-specific milling renderer under the same component responsibility principle. Its multichannel renderer starts from a train-only channel exemplar, adds the recipe's transient and band-energy structure, and preserves channel-specific location and scale. This renderer is calibrated inside the outer training fold, while Eqs. (4)–(7) define the bearing-channel model.

3.4. Verifier gates

Reliable pool construction requires evidence beyond recipe legality, so the verifier evaluates six complementary gate families:

1. **Legality:** exact shape, finite samples, non-degenerate channels, and absence of long repeated runs.
2. **Robust statistics:** class- and channel-conditional RMS, standard deviation, peak, kurtosis, skewness, and crest factor inside train-derived quantile or robust-union domains.
3. **Spectral shape:** triangular-band Welch-PSD fractions and PSD-CDF Wasserstein-1 distance to the class training reference.
4. **Envelope evidence:** resonance-band Hilbert demodulation, prominence near the requested BPFO, BPFI, BSF, or FTF frequency, harmonic support, and suppression of unsupported peaks for healthy candidates.
5. **Current evidence:** vector-current sideband scores become active when training data establish class separability; otherwise they remain audit-only measurements.

Algorithm 1 BREEZE recipe admission for one proposal slot**Require:** class y , condition m , LLM G , renderer R , verifier V , round limit K , renderer seeds S

```

1:  $q_{-1} \leftarrow \emptyset$ 
2: for  $t = 0$  to  $K$  do
3:    $r_t \leftarrow G(y, m, q_{t-1})$ 
4:   for all  $s \in S$  do
5:      $\tilde{x}_{t,s} \leftarrow R(r_t, m; s)$ 
6:      $(a_{t,s}, q_{t,s}) \leftarrow V(\tilde{x}_{t,s}, y, m)$ 
7:     if  $a_{t,s} = 1$  then
8:       archive recipe, seed, waveform hash, and full gate report
9:     return admitted slot and all diversity-admitted windows
10:    end if
11:  end for
12:   $q_t \leftarrow \text{AggregateFailures}(\{q_{t,s}\})$ 
13: end for
14: return rejected slot with terminal failure report

```

6. **Diversity:** a lower nearest-neighbour bound in the physical feature embedding relative to a train-derived real–real reference.

PU training data leave the MCSA score audit-only. Current-signature consistency is therefore reported only for pools whose training split activates that gate.

3.5. Closed-loop construction and audit records

Closed-loop admission must bound proposal cost while preserving every decision needed for audit and replay. Algorithm 1 states this contract for one proposal slot.

A *slot* is one requested class-conditioned recipe unit, whereas a *window* is the $C \times N$ tensor supplied to the classifier. One slot can archive candidates from several feedback rounds, and its admitted recipe can be rendered with several deterministic seeds. The frozen pipeline renders three seeds per recipe and can retain additional diversity-admitted expansions, so one admitted slot can yield multiple windows. This distinction matters because slot counts measure LLM proposal cost, while window counts measure the classifier’s augmentation budget.

Generation provenance determines which stages can be replayed exactly. The CWRU archive records the OpenAI-compatible model identifier `mimo-v2.5`, temperature 0.8, top- p 0.9, and maximum 900 output tokens, although the provider did not expose sampling seeds. The earlier PU archive preserves prompts, recipes, rendered pools, and local seeds; its original provenance record omits the exact provider version and top- p . Consequently, the frozen PU pools are reproducible from archived recipes and renderer seeds, while exact regeneration of the original language responses remains unavailable. Manuscript assembly required no new LLM call.

4. Experimental design

4.1. Public datasets and leakage-resistant splits

Leakage-resistant evaluation requires splitting acquisition units before sampling windows. The formal evidence uses the established PU and CWRU bearing benchmarks (Lessmeier, Kimotho, Zimmer and Sextro, 2016; Smith and Randall, 2015) and Berkeley milling from the NASA prognostics repository (Agogino and Goebel, 2007). Their file-, load-, or case/run-level protocols appear in Table 2, and only these grouped splits support formal claims.

The PU protocol isolates measurement files within each bearing. Its healthy, OR, and IR classes contain 1200, 1202, and 1444 training windows, together with 300, 301, and 361 test windows. The input retains one vibration and two phase-current channels.

The CWRU protocol tests both within-load diagnosis and source-load transfer. It evaluates a within-load0 split and three folds that hold out load1, load2, or load3. The frozen synthetic pools derive from load0, which belongs to the training side of these three folds. Reusing these pools gives the archived held-out-load0 fold target-load provenance, so that fold remains an audit artifact.

Table 2

Formal datasets and frozen protocols. Counts are windows after the declared split.

Dataset	Sampling and input	Classes	Operating regime	Train/test	Split unit
PU	8 kHz, 3 channels, 2048 samples	healthy, OR, IR	N09_M07_F10: 900 rpm, 0.7 Nm, 1000 N	3846/962	chronological measurement files within bearing; zero shared file/window keys
CWRU	12 kHz drive-end vibration, 2048 samples	healthy, IR, ball, OR	motor loads 0–3; defects 0.007–0.028 in	696/306 for within-load0	chronological within-file for load0; loads 1–3 held out separately for clean transfer tests
Berkeley	250 Hz, 6 channels, 512 samples	healthy ($VB < 0.2$), degraded ($VB \geq 0.2$)	feed, depth, and material combinations	1836/646	disjoint machining case/run groups

The Berkeley protocol isolates machining case/run groups before windowing. Its six channels contain spindle AC and DC current, table and spindle vibration, and table and spindle acoustic emission. Train-only inner validation fixed the healthy/degraded threshold and split before the one-time 40-seed formal run.

4.2. Recipe-source and augmentation controls

Testing the LLM contribution requires changing the proposal source while holding every downstream factor fixed. The central ablation therefore retains the applicable renderer, data split, synthetic budget, CNN, and paired seeds:

- **LLM + verifier:** class- and condition-aware JSON recipe, physical admission, and bounded feedback.
- **Rule + verifier:** engineer-written class template with train-supported parameter ranges, the same renderer, and the same verifier; no LLM call.
- **Random open-loop:** random legal recipe values rendered without feedback. On PU this is the matched non-semantic downstream control; a separate random-recipe verifier audit is also reported.
- **Noise augmentation:** real few-shot windows with channel-scaled jitter and amplitude scaling, using the same synthetic count.
- **Real only:** the paired few-shot subset without synthetic data.

The completed formal families match these controls to each dataset. CWRU compares LLM with rule, noise, and real only over within-load0 and held-out loads 1–3. Berkeley adds random open-loop, while PU Phase-A focuses on LLM versus rule and random open-loop.

The available baseline archive fixes the quantitative comparison boundary. Checkpointed TimeGAN and 1-D DDPM implementations reached smoke/development output before a completed predeclared formal run, which excludes them from the quantitative claims. Superiority to trained GAN or diffusion methods therefore awaits a matched formal run. FID/KID-style learned distances also remain unreported because the protocol lacks a fixed, validated domain feature extractor; their general role is described by Heusel, Ramsauer, Unterthiner, Nessler and Hochreiter (2017); Bińkowski, Sutherland, Arbel and Gretton (2018).

4.3. Few-shot classifier and statistical protocol

Matched downstream evaluation requires a fixed classifier and paired sources of randomness. All methods within a dataset therefore share the same compact 1-D CNN. Its three convolution blocks use 16, 32, and 64 channels, kernels 15, 9, and 5, stride two, max-pooling after the first two blocks, adaptive average pooling to length eight, and a linear classifier. Adam uses learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 32, and cross-entropy loss. PU uses 60 epochs; the frozen CWRU and Berkeley protocols use 20. Training-channel mean and standard deviation normalize both train and test data.

Table 3

PU Phase-A v2 file-split results (20 seeds; $n_{\text{syn}} = 150$ per class). Holm q values test LLM $K = 3$ superiority within each registered family.

n_{real}	metric	LLM $K = 3$	rule	random	q_{rule}	q_{random}
5	Acc.	0.777 ± 0.027	0.696 ± 0.053	0.450 ± 0.029	<0.001	<0.001
5	Macro-F1	0.776 ± 0.026	0.691 ± 0.049	0.436 ± 0.053	<0.001	<0.001
10	Acc.	0.796 ± 0.030	0.724 ± 0.042	0.483 ± 0.013	<0.001	<0.001
10	Macro-F1	0.800 ± 0.029	0.724 ± 0.040	0.488 ± 0.015	<0.001	<0.001
25	Acc.	0.818 ± 0.021	0.769 ± 0.025	0.529 ± 0.022	<0.001	<0.001
25	Macro-F1	0.821 ± 0.022	0.770 ± 0.024	0.527 ± 0.017	<0.001	<0.001

The dataset-specific budgets define the registered comparison cells. PU uses $n_{\text{real}} \in \{5, 10, 25\}$, 150 synthetic windows per class, and 20 paired seeds. CWRU uses the same shot counts, 20 synthetic windows per class, and 40 seeds. Berkeley uses $n_{\text{real}} \in \{2, 5, 10\}$, 20 synthetic windows per class, and 40 seeds. Accuracy and Macro-F1 are primary, while each paired seed selects the same real subset and classifier initialization for every method in its cell.

Registered inference preserves the pairing used in training. One-sided Wilcoxon signed-rank tests compare paired seeds (Wilcoxon, 1945), and Holm correction controls family-wise error within each predeclared comparison family (Holm, 1979). Benjamini–Hochberg values remain secondary audit output, while the registered decision follows Holm (Benjamini and Hochberg, 1995). We report direction, adjusted decision, seed count, and the complete pass pattern. An unadjusted p value near 0.05 remains descriptive; significance requires the adjusted Holm decision.

4.4. Physical and distribution diagnostics

Physical and distribution diagnostics complement downstream accuracy by measuring how each class relates to its outer-training reference. They include RMS, kurtosis, skewness, crest factor, peak value, BPFO/BPFI/BSF/FTF peak alignment where applicable, envelope prominence and harmonic support, PSD-CDF Wasserstein-1 distance, relative band-energy error, and nearest-neighbour distance. Wasserstein distance follows the optimal-transport interpretation of Villani (2009). The archived feature audit retains MMD with an RBF kernel and median pairwise-distance bandwidth (Gretton, Borgwardt, Rasch, Schölkopf and Smola, 2012), while the main figure reports directly interpretable distances. The core evidence uses these direct diagnostics; t-SNE, UMAP, and Mixup remain outside the frozen formal comparison (Zhang, Cisse, Dauphin and Lopez-Paz, 2018).

5. Results

5.1. LLM recipes outperform rule and random sources on PU

LLM recipes outperform rule and random open-loop recipes at every registered PU shot count for both diagnostic metrics. All 12 pairwise tests pass Holm correction ($q < 0.001$), and Table 3 reports the paired protocol means and standard deviations.

Seed-level deltas confirm that the PU LLM advantage is positive across every registered source comparison for Accuracy and Macro-F1. Figure 3 shows these paired deltas alongside their distributions. The matched design attributes this contribution to recipe selection within the complete BREEZE pipeline; the renderer and verifier retain responsibility for waveform construction and admission.

5.2. Load0-derived LLM recipes outperform comparators within load0 and on held-out loads 1–3

Load0-derived LLM recipes outperform every registered comparator within load0 and on provenance-valid held-out loads 1–3. The within-load0 means exceed rule and noise augmentation at each shown shot count (Table 4), while the combined protocol yields 72/72 positive Holm-corrected LLM–comparator decisions. Figure 9 reports the source-load0 LLM–rule transfer deltas: Accuracy improves by 1.7 to 3.5 percentage points, and Macro-F1 by 1.9 to 3.6 points across shots and held-out loads 1–3.

This result establishes source-load0 cross-load diagnostic utility for the tested held-out loads 1–3. A strict four-fold LOLO and cross-bearing scope remain outside this result because the reused synthetic pool gives the archived held-out-load0 fold target-load provenance.

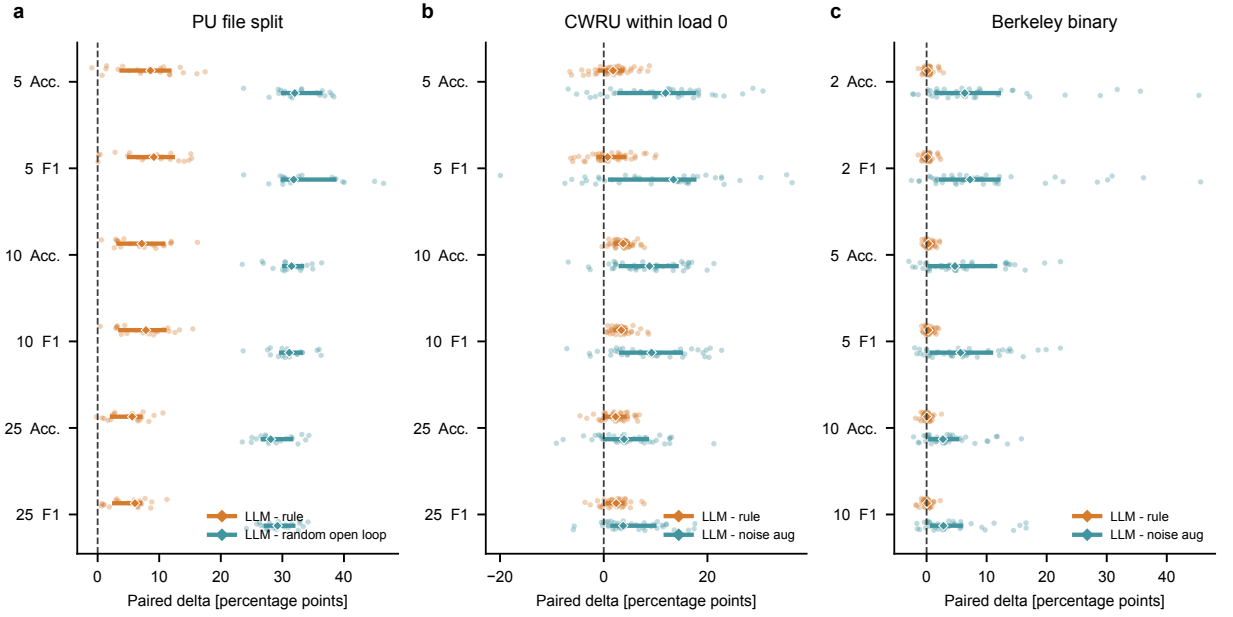


Figure 3: Paired downstream deltas. Points are seed-level differences; thick segments show interquartile ranges and diamonds show medians. (a) PU LLM minus rule or random open-loop. (b) CWRU within-load0 LLM minus rule or noise augmentation. (c) Berkeley LLM minus rule or noise augmentation. The dashed line is zero.

Table 4

CWRU within-load0 results (40 seeds). All 72/72 provenance-valid LLM-comparator tests across within-load0 and held-out loads 1–3 pass Holm correction. The archived held-out-load0 fold is excluded because it reuses a load0-derived synthetic pool.

n_{real}	metric	LLM	rule	noise augmentation
5	Acc.	0.564 ± 0.068	0.549 ± 0.053	0.454 ± 0.066
5	Macro-F1	0.567 ± 0.075	0.554 ± 0.062	0.460 ± 0.084
10	Acc.	0.621 ± 0.037	0.584 ± 0.033	0.534 ± 0.065
10	Macro-F1	0.646 ± 0.050	0.611 ± 0.046	0.554 ± 0.082
25	Acc.	0.674 ± 0.037	0.652 ± 0.037	0.628 ± 0.055
25	Macro-F1	0.711 ± 0.030	0.689 ± 0.035	0.655 ± 0.053

5.3. LLM recipes beat non-structured Berkeley controls and provide a qualified lower-shot advantage over rules

LLM recipes beat the non-structured Berkeley controls in all 12 registered comparisons. Against the strong structured rule reference, Accuracy passes at $n_{\text{real}} = 2$, while the two-shot Macro-F1 comparison remains outside the Holm decision; both metrics pass at $n_{\text{real}} = 5$. At $n_{\text{real}} = 10$, the LLM and rule results converge and both registered comparisons fall outside the Holm decision. This complete pattern yields 15 of 18 passes, with means reported in Table 5.

The Berkeley pattern localizes the additional LLM contribution to lower-shot settings. The rule recipe supplies a demanding structured reference, and its convergence with the LLM at ten shots shows that the additional LLM advantage narrows as the real sample count increases. The registered table supports this qualified source comparison within the binary Berkeley protocol, outside a generic milling state-of-the-art scope.

Table 5

Berkeley binary milling formal test (40 seeds; $n_{\text{syn}} = 20$ per class). This is a partial result: all comparisons against noise augmentation and random open-loop pass, whereas LLM versus rule passes only Acc. at $n_{\text{real}} = 2$ and both metrics at $n_{\text{real}} = 5$.

n_{real}	metric	LLM	rule	noise aug.	random open-loop
2	Acc.	0.786 ± 0.021	0.783 ± 0.022	0.695 ± 0.110	0.524 ± 0.147
2	Macro-F1	0.778 ± 0.018	0.776 ± 0.019	0.675 ± 0.112	0.450 ± 0.127
5	Acc.	0.784 ± 0.030	0.779 ± 0.031	0.724 ± 0.068	0.614 ± 0.097
5	Macro-F1	0.773 ± 0.031	0.769 ± 0.032	0.709 ± 0.071	0.514 ± 0.106
10	Acc.	0.787 ± 0.024	0.786 ± 0.024	0.750 ± 0.049	0.654 ± 0.062
10	Macro-F1	0.777 ± 0.024	0.776 ± 0.023	0.734 ± 0.056	0.572 ± 0.081

Table 6

Class-averaged PU physical diagnostics for fixed 150-window/class pools. Distances and errors are reference-relative diagnostics, not a downstream ranking; diversity has no preferred direction.

pool	RMS W1	PSD W1	band-energy error	NN diversity
llm	0.008	236.9	0.044	17.10
rule	0.031	271.6	0.235	14.20
random_open_loop	0.024	516.1	0.749	52.50
noise_aug	0.004	309.1	0.014	46.09

5.4. PU diagnostics show complementary physical agreement and non-copy behaviour

PU diagnostics show that admitted LLM pools combine class-relevant waveform evidence with non-duplicate nearest-neighbour distances. Under the same fixed demodulation procedure, Fig. 4 compares real, LLM, rule, and random windows. Admitted LLM and rule recipes preserve impulsive and envelope structure more consistently than the random control while retaining visible stochastic variation. These traces illustrate the population-level distributions in Figs. 5–6.

Class-averaged PU distances favour LLM recipes over the matched recipe sources on PSD, band energy, and RMS. Table 6 reports lower LLM PSD distance and band-energy error than rule and random open-loop, together with lower RMS distance than both recipe sources. Noise augmentation is closest in RMS and band energy because it directly perturbs real windows. The LLM nearest-neighbour distance also separates the pool from its closest real reference under the frozen feature metric. This diversity value requires joint interpretation with fidelity because arbitrarily large distances can indicate unrealistic samples.

External downstream tests, matched source controls, train/test separation, and failed cross-condition experiments provide evidence beyond the shared physical features. The prompt, renderer, and verifier encode some of the same bearing quantities, which makes gate compliance necessary but insufficient evidence of real-world physics.

5.5. Closed-loop admission exposes proposal yield and class-specific failure mechanisms

Closed-loop accounting separates proposal cost from the downstream augmentation budget. PU requests 150 proposal slots per class; Fig. 7(a) reports archived round candidates, panel (b) separates proposal slots, admitted slots, and kept rendered windows, and panel (c) compares recipe sources. At $K = 3$, the LLM process admits 60%, 60%, and 71% of healthy/OR/IR slots.

Recipe source determines whether the verifier can construct a balanced pool. The random-recipe audit admits no slot, whereas rule admission varies by class. This contrast makes the fixed rule a meaningful but incomplete substitute for LLM proposal under the PU verifier.

Terminal failures reveal class- and source-specific rejection mechanisms. Figure 8 reports the non-exclusive rates. Random recipes fail statistical and spectral gates much more often than cached LLM recipes. Healthy LLM candidates are rejected most often for unsupported envelope evidence, whereas fault candidates more often violate the robust statistical domain. A slot may fail several gates, so the percentages are non-additive. The same report supplies the next feedback round with each measured violation and its bound.

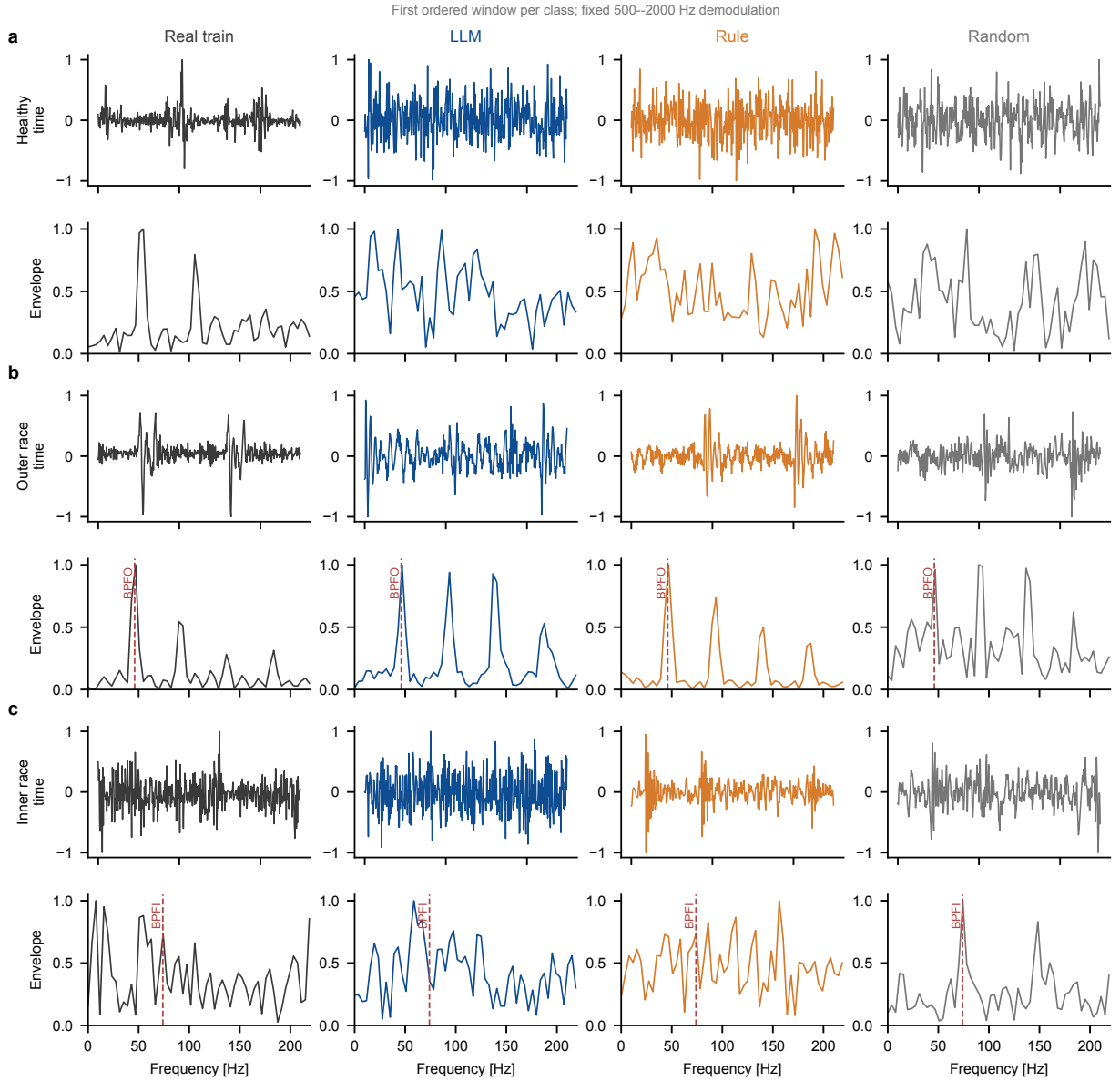


Figure 4: PU waveform and envelope-spectrum comparison. Rows are healthy, outer-race, and inner-race classes; columns compare real training, LLM, rule, and random open-loop windows. All envelope spectra use the same fixed 500–2000 Hz demodulation band. Vertical markers denote the class-relevant kinematic frequency.

6. Cross-condition evidence and negative results

These failures define the method’s applicability, and their protocol value is to prevent unsupported extrapolation from positive within-condition results.

6.1. CWRU load transfer succeeds while PU LOCO remains a failed stress test

CWRU source-load0 recipes transfer to the tested loads 1–3, whereas the PU LOCO protocol remains unsuccessful. Figure 9 places these outcomes side by side: PU v1 fails 57/96 registered cells, and v2 fails 60/96. The PU failures

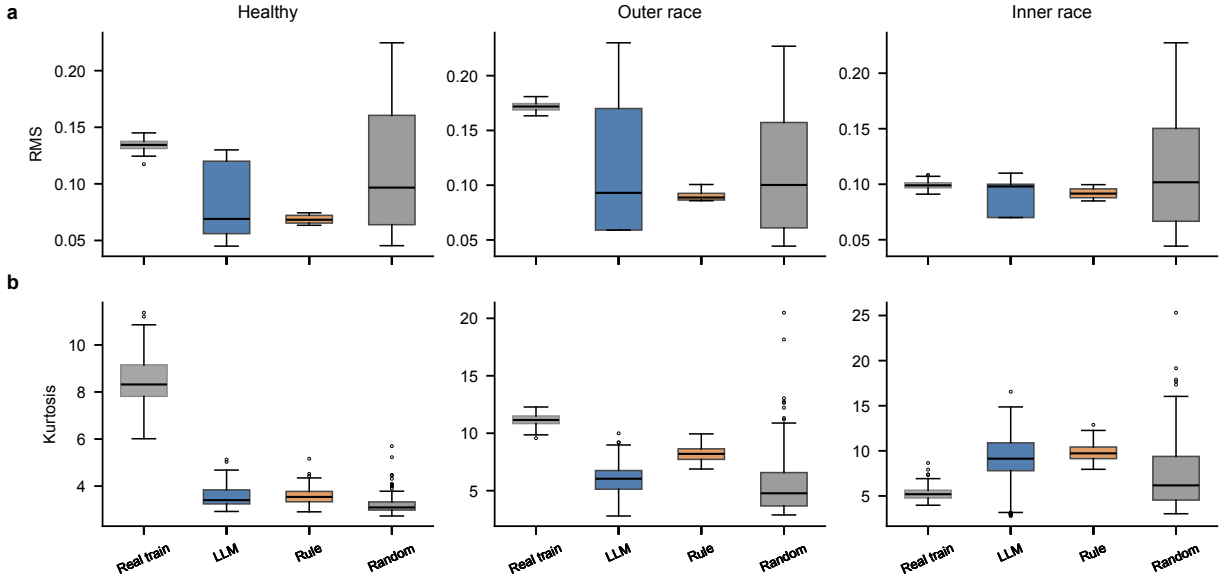


Figure 5: Class-conditional RMS and kurtosis distributions for the real PU training split and matched synthetic pools. Boxes show the frozen pool distributions; admission narrows unsupported statistical excursions without forcing equality to a single reference value.

cluster by held-out condition and persist across Accuracy, Macro-F1, and shot counts, locating the unresolved boundary in condition-dependent morphology.

The six-stage PU audit localizes this boundary from kinematics through source-only evidence (Fig. 10). v1 exposed a kinematic mismatch, and v2 corrected frequency handling while downstream performance remained unsuccessful. v3 stopped when train-condition morphology mapping could not form an admissible balanced pool. v4 stopped after both train-only candidate-pool strategies failed admission, and v5 showed that its candidate criterion also admitted wrong-label/noise controls. v6 stopped because source-only cyclostationary coherence failed to separate all four held-out targets. The formal held-out evaluations are v1 and v2; v3–v6 remain development or admission stops.

6.2. Milling boundaries

The core milling evidence is limited to datasets with complete, auditable protocol metadata. The private machine-tool dataset used in earlier drafts therefore leaves the core evidence. Its speed, geometry, file-to-operation manifest, and audited physical label mapping are incomplete, which prevents a public reproducible portability claim.

UMich and MU-TCM define two distinct protocol stops. In the available UMich protocol, experimental condition and process metadata are confounded with wear labels, allowing a classifier to exploit condition identity. The current files leave these effects entangled, so a valid formal test requires repeated conditions or a revised acquisition design. MU-TCM v3 stops at train-only inner validation, where only 2/6 predeclared comparisons pass; preregistration and formal held-out evaluation were therefore withheld. Both datasets remain boundary evidence.

7. Discussion

7.1. The three evaluation questions define a bounded LLM contribution

The source ablations answer the first evaluation question by establishing an LLM recipe contribution within the matched BREEZE pipeline. On PU, the random-recipe-plus-verifier control produces 0 admitted slots and fails to form a balanced pool, while random open-loop augmentation is substantially weaker downstream. LLM recipes also exceed the fixed rule source in every registered PU cell and every provenance-valid CWRU family.

The class-conditional diagnostics answer the second question with complementary train-supported physical evidence. Admitted pools exhibit waveform, envelope, PSD, band-energy, statistical-distance, and non-copy behaviour

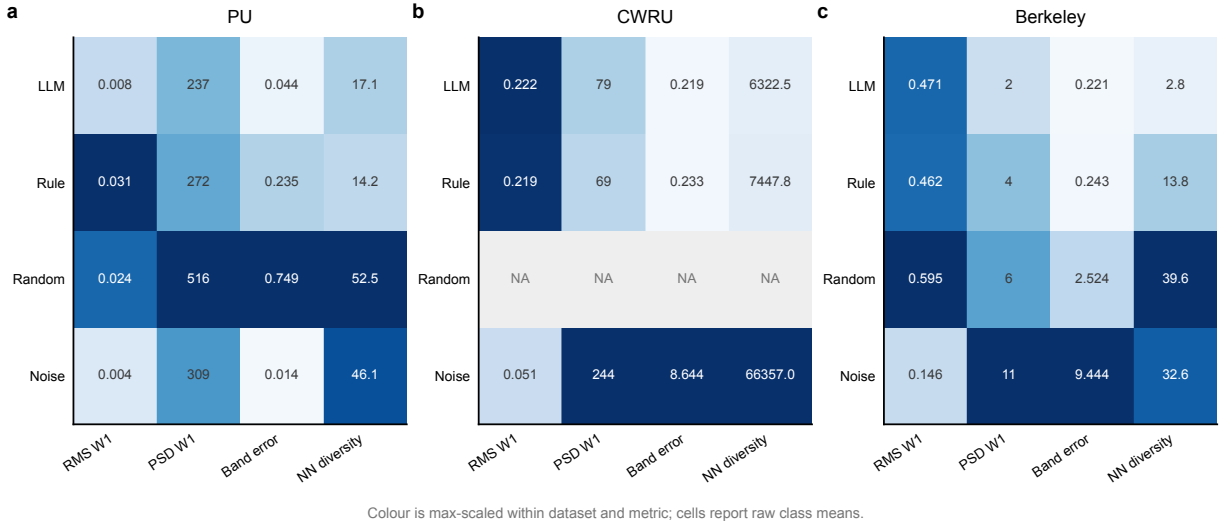


Figure 6: Class-averaged physical diagnostics on PU, CWRU, and Berkeley. Cells contain raw RMS Wasserstein-1, PSD-CDF Wasserstein-1, relative band-energy error, and nearest-neighbour diversity values; colour is scaled within each dataset and metric only. NA denotes a pool that does not exist in the frozen archive, not an imputed value. Diversity has no universally optimal direction.

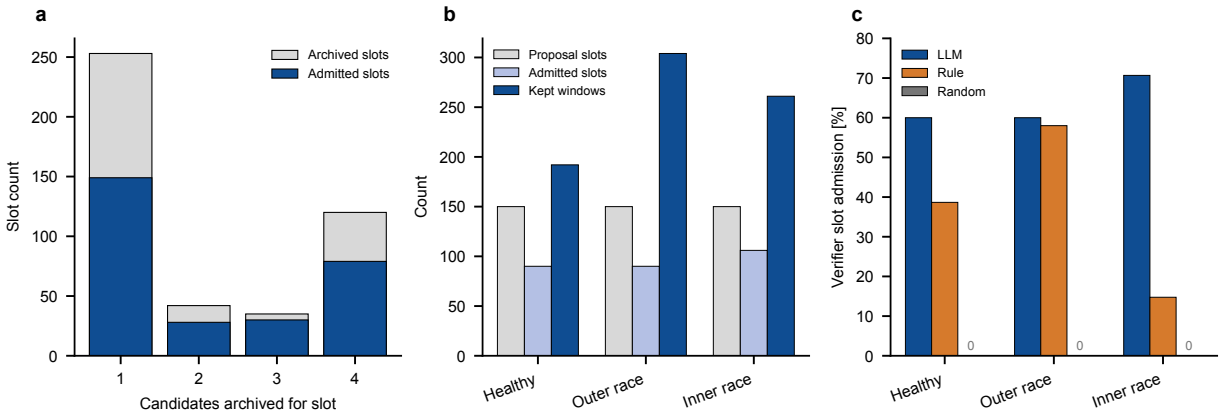


Figure 7: PU proposal accounting. (a) Number of archived candidates per proposal slot. (b) One-to-many mapping from 150 class slots to admitted slots and diversity-kept windows. (c) Verifier slot-admission rates for LLM, rule, and random recipes. Random open-loop remains a downstream control; random recipes subjected to the verifier produce no balanced admitted pool.

under the frozen metrics. These measurements support admission under the declared gates while leaving full physical truth outside their observational scope.

The downstream protocols answer the third question by locating both stable utility and its transfer boundary. LLM recipes improve the registered PU and provenance-valid CWRU comparisons and provide a qualified lower-shot Berkeley advantage. Their extra value converges with the rule recipe at ten Berkeley shots, while PU LOCO remains unsuccessful across six predeclared stages.

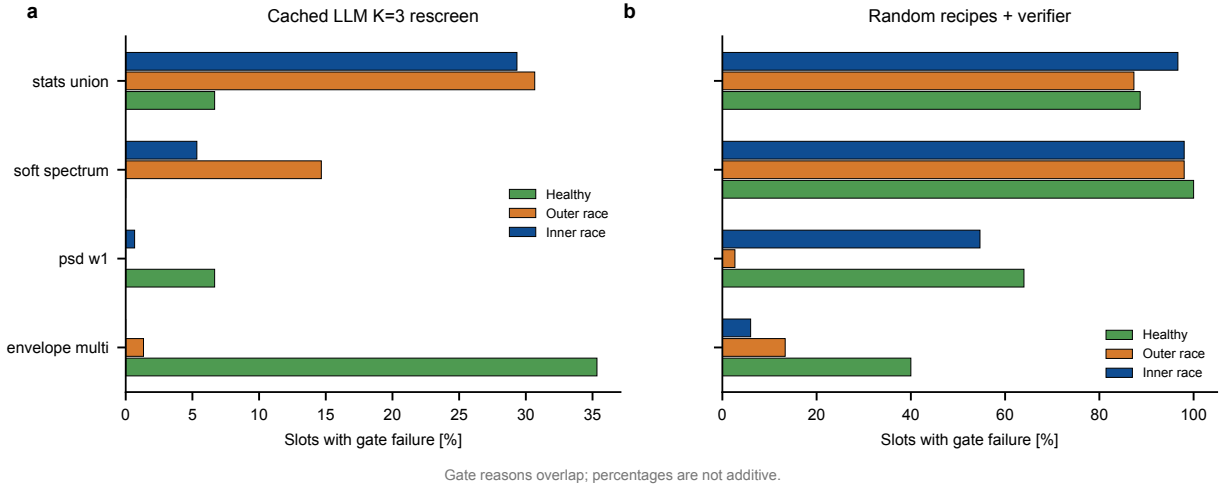


Figure 8: Non-exclusive PU gate-failure rates by class and recipe source. (a) Cached LLM $K = 3$ rescreen. (b) Random recipes passed through the same verifier. Statistics, soft-spectrum, PSD-W1, and multi-band envelope failures are computed from frozen slot reports.

Together, these answers position BREEZE as a train-calibrated control layer around a structured black-box proposal source. This operating point supports deployments that require inspectable admission records and bounded resampling without training a new target generator.

7.2. Physical admission evidence versus physical truth

Train-calibrated gates provide auditable physical evidence within a finite set of observables. A candidate can satisfy RMS, PSD, and envelope gates while missing load-path, transfer-function, non-stationary, or degradation mechanisms; a rare real sample can likewise fall outside a finite training boundary. BREEZE therefore reports class-wise distances, downstream tests, and negative transfer results alongside gate pass rates. A pass means “admissible under the declared training-calibrated gates,” not proof of membership in the full real distribution.

Matched source controls partially address the circularity created by shared renderer and verifier kinematics. Rule, random, and LLM sources all encounter the same renderer and physical checks, yet the LLM source consistently forms the useful admitted pools in the registered PU comparisons. Further reduction of circularity requires external expert scoring, independent physical observables absent from the prompt, and new condition-disjoint datasets.

7.3. Reproducibility and remaining blockers

The repository supports replay from archived recipes through reported statistics. It stores split manifests, prompts, recipe JSON, local renderer seeds, waveform hashes, per-gate reports, pool manifests, seed-level classifier rows, registered tests, and table-generation assertions. Long-running pipelines are append-only or checkpoint-resumable. Frozen CSV/NPZ artifacts generate every reported number, and NA explicitly marks an unavailable pool.

Two provenance and baseline blockers remain. The original PU archive omits the provider version and provider-side sampling seed, so exact replay begins from the archived recipes instead of the original LLM response. Completed formal TimeGAN/DDPM and additional diagnostic-backbone comparisons are also unavailable, which keeps state-of-the-art and universal classifier claims outside the supported scope. Future work should complete those matched baselines, repeat the protocol with at least one non-CNN diagnostic backbone, and preregister a new PU cross-condition representation before accessing held-out outcomes.

8. Conclusion

BREEZE separates LLM recipe proposal, fixed seeded rendering, train-calibrated physical admission, and downstream diagnosis. Across the frozen public protocols, LLM recipes outperform matched controls on PU and CWRU,

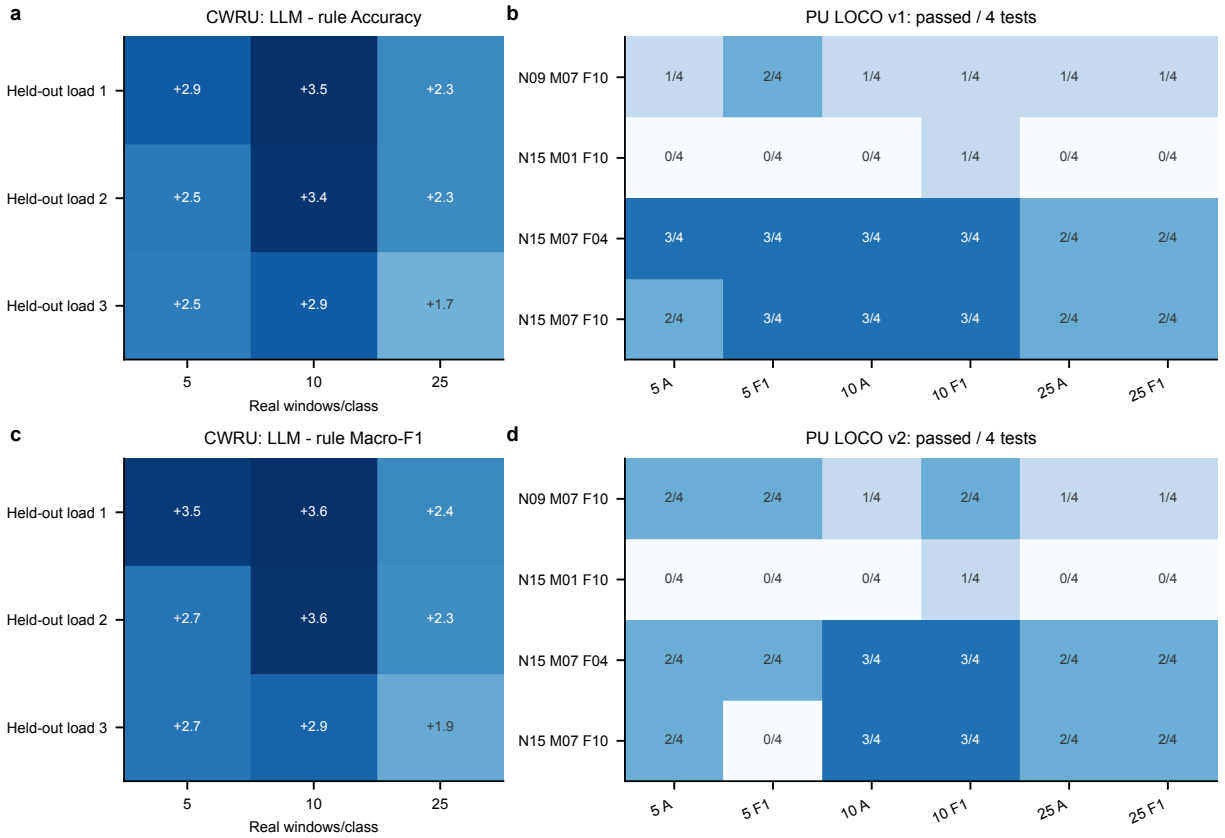


Figure 9: Cross-condition evidence. (a,c) CWRU LLM-rule Accuracy and Macro-F1 deltas when load0-derived pools are evaluated on held-out loads 1–3. The archived held-out-load0 fold is excluded for target-load provenance. (b,d) Number of passed PU LOCO tests out of four registered comparisons in each condition-shot-metric cell for v1 and v2.

provide a qualified lower-shot Berkeley benefit, and produce inspectable physical diagnostics and gate records. The supported scope is training-free, physics-verified admission for the stated few-shot tasks; zero-data diagnosis, formal guarantees, PU cross-condition generalization, and universal generator superiority remain outside it.

Data and code availability

The release package connects each reported result to public data and frozen evidence. PU and CWRU are public benchmarks, and Berkeley milling is available through the NASA prognostics repository subject to its terms. The script `scripts/build_paper_tables.py` generates the manuscript tables, while `analysis/evidence_ledger.md` and `analysis/claim_evidence_map.md` enumerate numerical sources, availability, and permitted wording. Release artifacts include gate thresholds, per-sample pass/fail records, pool manifests, split audits, and seed-level downstream outputs. API credentials and raw third-party datasets remain outside the release.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

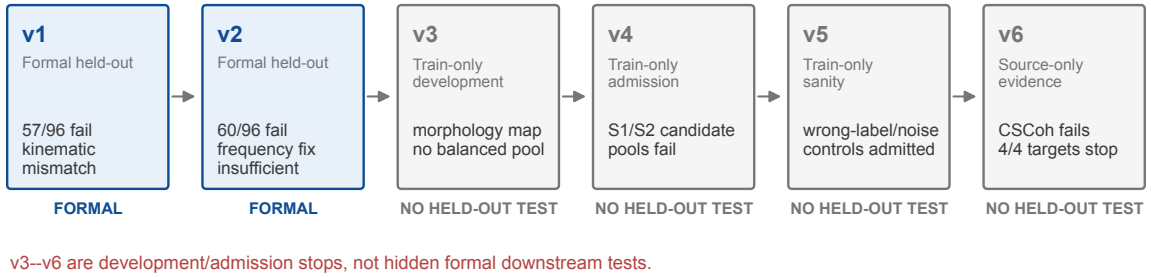
PU leave-one-condition-out: the complete predeclared failure chain

Figure 10: Complete PU LOCO failure chain. v1 and v2 are formal held-out evaluations; v3–v6 stop at train-only morphology, admission, sanity, or source-only evidence gates. The sequence prevents unsuccessful development stages from being reported as completed cross-condition tests.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI Codex for code inspection, experiment orchestration, figure and table assembly, and language editing. The authors reviewed the evidence, verified the reported results, and take responsibility for the manuscript.

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